WEEKLY JOURNAL DIGEST

Weekly journal digest for 2026-08-10 to 2026-08-16

Curated articles in political communication and computational social science.

SUMMARY

This week's COMAP Journal Bot brief is unusually rich in research on the politics of attention and institutional control. New work tracks how entertainment lowers resistance to authoritarian propaganda in China, how TikTok memes and digital populism organize political affect, and how media curation, judicial coverage, and cross-border news flows shape what audiences learn and notice.

Three clusters deserve particular attention:

- Authoritarian and multimodal communication: propaganda movies, campaign memes, platform-native political curation, and contested electoral legitimacy.
- AI and computational governance: agentic capability profiles, large language models and federal research funding, scalable multi-agent coordination, and interpretable behavioral measurement.
- Institutions, inequality, and policy: gerrymandering, migration and crime, climate commitments, public-sector commercialization, tariff psychology, and a substantial PNAS collection on illegal markets.

This digest was prepared early on Saturday, 2026-08-15, before the nominal window closes on Sunday, 2026-08-16. It reflects records available at collection time; publications appearing later in the window may be recovered in a subsequent catch-up or late-additions section.

HIGHLIGHTS

Entertainment as Trojan Horse: Voluntary Consumption of Propaganda in China

POLITICAL COMMUNICATION · 2026-08-13

Why it matters · Shows how personality and narrative entertainment make authoritarian propaganda more attractive-even to citizens least inclined to seek it out-and more effective as political persuasion.

Open article →

Kamala is Brat: TikTok and the Affective Politics of Memes

POLITICAL COMMUNICATION · 2026-08-12

Why it matters · Connects TikTok meme culture to campaign affect, offering a timely account of how platform-native visuals and vernacular participation shape political identification.

Open article →

Affective Legitimacy and Digital Populism: Evidence from Romania's Annulled Presidential Elections

POLITICAL COMMUNICATION · 2026-08-13

Why it matters · Uses an extraordinary electoral rupture to examine how digital populism converts affect into perceived legitimacy when formal democratic procedures are contested.

Open article →

Trending Topics, Growing Knowledge? How Curating Actors' Political Social Media Content Affects Youth's Political Knowledge

POLITICAL COMMUNICATION · 2026-08-10

Why it matters · Tests whether political content selected and circulated by social-media curators actually improves young audiences' knowledge, directly linking platform mediation to democratic learning.

Open article →

Uneven patterns of cross-border media coverage following natural disasters

NATURE HUMAN BEHAVIOUR · 2026-08-14

Why it matters · Maps which disasters travel across national news systems and which remain invisible, clarifying the geopolitical filters that structure international attention.

Open article →

Agentic profiles for effective AI governance

NATURE · 2026-08-12

Why it matters · Offers a governance framework centered on what increasingly autonomous AI systems can do, a useful bridge between technical capability profiles and accountable oversight.

Open article →

The rise of large language models and the direction and impact of US federal research funding

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-08-11

Why it matters · Links the LLM boom to shifts in federal research investment and downstream scientific impact, with direct implications for computational social-science infrastructure.

Open article →

AI agents can coordinate via majority-following beyond human scale

SCIENCE ADVANCES · 2026-08-14

Why it matters · Demonstrates scalable coordination among AI agents through simple social signals, raising concrete questions about collective behavior, robustness, and governance in multi-agent systems.

Open article →

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Previous Week Catch-Up

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- Journal of Communication
- Communication Research
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- Nature Human Behaviour
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- Proceedings of the National Academy of Sciences
- Science Advances
- Nature
- Science

COLLECTION SNAPSHOT

TOTAL COLLECTED CANDIDATES	NEW THIS WEEK CANDIDATES	PREVIOUS WEEK CATCH-UP CANDIDATES
323	161	161
LATE ADDITIONS CANDIDATES	**CURATED DIGEST BELOW**	**NOTE**
1	36 new, 23 catch-up, 0 late additions	These counts reflect the complete collected set before manual relevance filtering, while the attached PDF reflects the curated digest.

FULL CURATED DIGEST

New This Week

Political Communication

Affective Legitimacy and Digital Populism: Evidence from Romania's Annulled Presidential Elections

2026-08-13 · DOI 10.1080/10584609.2026.2714818

Antonio Amuza; Vasile Dincu

University of Bucharest

Abstract. On 6 December 2024, Romania's Constitutional Court annulled the first round of presidential elections, triggering an exceptional crisis of democratic legitimacy in post-communist Europe. While procedurally justified, the decision sparked intense public reactions, both emotional and political. The findings confirm that populist interpretations of institutional actions proliferate in fragmented and emotionalized digital media environments, where distrust is personalized and intensified. Drawing on a nationally representative survey (N = 964), conducted by IRES shortly after the annulment, the study examines how Romanian citizens perceived the legitimacy of the Court's decision and identifies the key socio-political and media-related factors shaping these perceptions. Binary logistic regression and mediation models were used to assess the drivers of perceived illegitimacy. TikTok use, negative emotions (anger, disappointment, resentment), political distrust, and anti-establishment party support significantly increase the likelihood of viewing the decision as illegitimate, whereas trust in journalists reduces it. The study contributes to the literature on populism as a communicative style and emotional register, showing how legitimacy itself becomes a contested symbol in digital environments marked by personalization, fragmentation, and affective intensification. We argue that institutional legitimacy is no longer grounded solely in procedural legality but is increasingly shaped by emotionally resonant narratives circulating through digital platforms. We complement the survey with a headline-level media analysis (N = 3,923; Dec 2024 - Aug 2025), to serve as a contextual counterfactual: instead of escalating antagonism, mainstream media gradually stabilize institutional language. A keyness comparison shows that the Early window (first 14 days) is driven by institutional/legal terms (CCR, decision, annulment), while the Late window becomes actor-centric (Simion, Antonescu, Ponta, Bolojan). Over time, lexical entropy declines markedly, signaling a standardization of vocabulary around procedural frames rather than escalating antagonism. Taken together, the results point to a decoupling between media normalization and individual affect, suggesting that democratic legitimacy in platformized communication environments is increasingly produced through affective legitimacy rather than procedural authority.

Read article →

Entertainment as Trojan Horse: Voluntary Consumption of Propaganda in China

2026-08-13 · DOI 10.1080/10584609.2026.2701201

Hanzhang Liu; Linan Yao

Florida State University; Vanderbilt University

Abstract. For authoritarian propaganda to change hearts and minds, it must first reach the masses and capture their attention. In an era of media proliferation, what can an authoritarian regime do to make ordinary citizens consume pro-regime propaganda voluntarily? How can it lower citizens' resistance to being indoctrinated? In this article, we answer these questions by studying a new generation of propaganda movies in contemporary China. Leveraging observational data and a pre-registered survey experiment, we find that entertainment elements in propaganda movies - both personality-based and narrative-based - play a key role in appealing to viewers and increasing consumption; this effect is especially pronounced among groups that are the least inclined to expose themselves to propaganda in the first place. Also, viewers are more receptive to political indoctrination when it is interwoven with entertainment, even when they recognize such content.

Combined, these findings highlight the importance of entertainment in authoritarian propaganda: not only does it serve as a vehicle to disseminate political indoctrination, but it also functions as a source of appeal that renders such indoctrination more palatable to the masses.

Read article →

Kamala is Brat: TikTok and the Affective Politics of Memes

2026-08-12 · DOI 10.1080/10584609.2026.2715697

Laura Cervi; Sarah J. Jackson

Autonomous University of Barcelona; University of Pennsylvania

Abstract. In this article, we examine TikTok memes about former Vice president Kamala Harris that circulated through the digital registers of the Charli XCX-inspired "Kamala is Brat" trend. By analyzing the form, function, and affective resonance of these memes, we explore how creators interpreted and reframed Harris's image leading up to the 2024 U.S. presidential election. Using a multimodal analysis, we introduce a three-layer analytic that distinguishes memetic form, communicative function, and affective character. Our findings show how creators mobilized recognizable meme templates, narrative positioning, and affective cues to recast Harris through identity-signaling and aesthetic contrast to her opponent. We also find that TikTok's features enabled a pop culture intertextuality in which other viral trends - including the appeal of "it girl" Julia Fox, and the symbolic ambiguity of signifiers ranging from Pulp Fiction, The Matrix, and coconuts - become linked to the candidate. We argue that platform-native meme cultures on TikTok operate as mechanisms of vernacular electoral mediation, enabling creators to reinterpret candidate imagery and political narratives through the aesthetic conventions of platform culture. In doing so, this work demonstrates how participatory meme practices contribute to the symbolic construction of electoral candidates in contemporary digital media environments.

Read article →

Policing the Political Boundary: How the r/BarstoolSports Audience Confronts Politicization

2026-08-12 · DOI 10.1080/10584609.2026.2715700

Elliott Edsall

University of Minnesota - Twin Cities

Abstract. Recent scholarship attends to politicized popular cultural spaces. Scarcely understood is how audiences of these spaces contend with this scrutiny. In response, I introduce the concept of anti-political space to describe how fans of politicized media spatially distinguish popular cultural spaces from institutional politics. This is conceptualized through a critical discourse analysis of the official Barstool Sports (Barstool) subreddit from 2019 to 2025, where fans share critical news coverage of Barstool. Across 16 analyzed threads, criticism is rendered unfounded or demonstrative of politically obsessed liberal culture. I argue fans' varied expressions of victimhood and estrangement, while nominally defending against politicization, showcase how reactionary political discourse appears beyond institutional political spaces. Theoretically, this analysis recognizes how symbolic contests over what constitutes politics occur through boundary work. As the political relevance of popular cultural spaces increases in the contemporary media landscape, this analysis underscores how boundary work defends against the felt intrusion of politics into popular culture. Amid broader sociotechnical changes, I suggest anti-political spaces are poised to be a continued feature of the contemporary media environment.

Read article →

Examining the Influence of Media Coverage on Information-Seeking of Judicial Candidates

2026-08-10 · DOI 10.1080/10584609.2026.2716378

Marcy Shieh

University of Wisconsin-Whitewater

Abstract. Abstract unavailable.

Read article →

Trending Topics, Growing Knowledge? How Curating Actors' Political Social Media Content Affects Youth's Political Knowledge

2026-08-10 · DOI 10.1080/10584609.2026.2708127

Margot Lissens; Cato Waeterloos; Laura Vandenbosch; Desiree Schmuck

KU Leuven; University of Vienna

Abstract. Whether youth can gain political knowledge through social media is long debated. Existing research has insufficiently accounted for different curating actors of political information, an indirect learning logic, and within-person changes. To fill these gaps, we conducted a large-scale three-wave panel study among 16- to 30-year-olds (NT3 = 2276) to examine how exposure to different social media actors such as politicians, peers, or influencers relates to the acquisition of current affairs knowledge over time. We investigated perceived opinion leadership gratifications as a novel underlying mechanism in this process. While increased exposure to all curating actors' political content was related to higher opinion leadership gratifications in the same wave, we found little indirect learning effects over time on the within-person level through obtained gratifications. This association emerged only during the national election period, when perceived opinion leadership gratifications from peers' political content were linked to above-usual gains in objective political knowledge over time.

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American Journal of Political Science

(Non)belonging and (un)freedom

2026-08-13 · DOI 10.1111/AJPS.70081

Yuna Blajer de la Garza

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Abstract. Abstract This article centers on the concept of "belonging" to argue that belonging and freedom are fundamentally related, and that this relationship cannot be easily captured in formal terms alone. I contend that experiences of incomplete belonging among citizens deserve attention from political science because, in addition to their more obvious challenge to equality, they also erode freedom, even in the context of robust formal guarantees. Citizens perceived as not fully belonging are more likely to eschew behaviors or opinions they know can exacerbate the likelihood of being perceived as interlopers. But although this challenge to individual freedom may be by itself worrisome, I focus on the ways in which some citizens' unequal belonging undermines collective freedom-what Hannah Arendt calls political freedom-by confining some people's sincere expressions to narrow public spaces, limiting pluralism, and depleting the public space of opportunities for deliberation and political action.

Read article →

Global harms, local profits: How the uneven costs of natural disasters affect support for green political platforms

2026-08-13 · DOI 10.1111/AJPS.70078

Silvia Pianta; Paula Rettl

Assistant Professor, Business, Government, and the International Economy Unit Harvard Business School Boston Massachusetts USA; CMCC Foundation-EuroMediterranean Center on Climate Change Lecce Italy; Junior Scientist, RFFCMCC European Institute on Economics and the Environment Milan Italy

Abstract. Abstract Largescale fires are becoming increasingly common due to climate change. While conventional wisdom suggests that firsthand experiences with natural disasters foster green coalitions by raising awareness of environmental degradation, we propose an alternative view: Natural disasters can heighten the perceived tradeoff between economic development and environmental policies. This study examines the case of Brazil, where largescale fires, by clearing natural vegetation, create new opportunities for land grabbing and the expansion of agriculture and livestock production. Using satellite, administrative, and electoral data, we provide evidence that largescale fires increase support for green candidates only in municipalities that are unlikely to reap economic benefits from fires. Survey data further indicate that voters prioritize environmental protection over economic growth only when they stand to receive limited or no benefit from the newly cleared land. Our findings suggest that the potential for natural disasters to foster green coalitions is contingent upon distributive concerns.

Read article →

Durable majority gerrymanders: Where partisan gerrymandering can displace democracy

2026-08-10 · DOI 10.1111/AJPS.70074

Maxwell Palmer; Benjamin Schneer

Associate Professor, Department of Political Science Boston University Boston Massachusetts USA; Associate Professor, Harvard Kennedy School Cambridge Massachusetts USA

Abstract. Abstract We develop the concept of, and estimation tools for, durable majority gerrymanders : electoral district maps drawn to reduce an opposing political party's probability of winning a majority in a legislative chamber. Directly interpretable, forwardlooking, and motivated by the democratic principle that a party in power should have some chance of losing it, this measure provides new insights into the role of redistricting in state legislative elections. We show that, when map drawers are unconstrained, the ability to create durable majorities is so widespread that at least one party in every state can draw a map where a majority of legislative districts withstand almost any likely future electoral swing. Enacted maps are less durable due to a combination of underlying geography, voter partisanship, and statelevel guidelines. This paper provides the theoretical framework and empirical tools to understand which gerrymandered maps enable statelevel majorities to emerge and to endure.

Read article →

Endogenous coalition formation in policy debates

2026-08-10 · DOI 10.1111/AJPS.70079

Philip Leifeld; Laurence Brandenberger

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Abstract. Abstract Political actors form coalitions around their joint normative beliefs in order to influence the policy process on contentious issues such as climate change or population aging. Policy process theory maintains that learning within and across coalitions is a central predictor of policy change but has yet to explain how policy learning works. This article explains the formation and maintenance of coalitions by focusing on the ways actors adopt policy beliefs from other actors in policy debates. A policy debate is a complex social system in which temporal network dependence guides how actors contribute ideological statements. Belief adoption matters in three complementary ways: bonding , which exploits cues within coalitions; bridging , which explores new beliefs outside one's perimeter; and repulsion , which reinforces polarization between coalitions and cements their belief systems. We formalize this theory of endogenous coalition formation in policy debates and test it on a microlevel empirical data set.

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Nature Human Behaviour

Uneven patterns of cross-border media coverage following natural disasters

2026-08-14 · DOI 10.1038/S41562-026-02512-6

Thiemo Fetzer; Prashant Garg

Abstract. Abstract Many natural disasters central to climate-policy debates are hydro-meteorological hazards, yet it remains unclear how media attention is distributed across disaster types. Using 135 million news articles from 466 sources in 123 countries since 2016, we estimated cross-border reporting changes after disasters. Attention rises but is highly uneven (95% confidence intervals are shown in parentheses after the estimates): short-run increases are largest following earthquakes (= 0.0785 (0.0758, 0.0812)), dry-mass movements (= 0.0531 (0.0349, 0.0713)) and volcanic eruptions (= 0.0425 (0.0359, 0.0490)), and smaller for floods (= 0.0069 (0.0062, 0.0076)) and droughts (= 0.0001 (0.0036, 0.0039)). Conditional on severity and duration, hydro-meteorological disasters receive less attention than geophysical disasters (= 0.0065 (0.0115, 0.0015)). Coverage is higher for 100+ versus 0-9 deaths (= 0.0360 (0.0231, 0.0490)), with stronger fatality gradients for country pairs with tighter social ties (= 0.1498 (0.0822, 0.2174)) and ancestral relatedness (= 0.2965 (0.1887, 0.4043)).

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Homophily in grant evaluations

2026-08-13 · DOI 10.1038/S41562-026-02554-W

Nicola Fuchs-Schundeln; Felix Holub

Abstract. Abstract Funding agencies play a central role in science by awarding grants to researchers. Grant review can be influenced by bias, including preferences for applicants who share reviewers' gender, country or citizenship. Using 37,073 applications for European Research Council grants with multiple reviews per proposal, we show that reviewers who share an applicant's country of work or citizenship award higher scores than reviewers with no such match. This effect is strongest when both country and citizenship align. Shared gender increases scores to a smaller extent. Further analyses indicate that these findings are unlikely to be driven by shared tastes or informational advantages. At the panel level, correlational evidence shows that applicants are more likely to advance past the first stage when the chair or more panel members share their country of work or citizenship. Country of work and citizenship may therefore be relevant sources of bias in international grant review.

Read article →

Nature Communications

Public institutions for cleantech commercialization in the United States

2026-08-13 · DOI 10.1038/S41467-026-76484-6

Em Finkelstine; Laura Diaz Anadon; Jonas Meckling

Abstract. Abstract The United States has been building public institutions to promote cleantech commercialization since the 1970s oil crisis. The ascent of new industrial policy has accelerated this process substantially since 2020, resulting in a doubling of the total number of institutions by 2024. Introducing a dataset of 122 institutions, we show how institutions perform five functions: goal setting, coordination, derisking, imposing discipline, and external engagement. Further, we demonstrate how post-2020 institutions extend and diverge from historical trends, highlighting the recent buildout of planning capacity, the widespread use of conditionalities, and the pivot toward engagement with local and non-firm actors. We conclude with an analysis of how the second Trump administration is transforming the cleantech commercialization system through retrenching, redesigning, repurposing, and maintaining institutions. We offer a bottom-up, institution-level examination of the emergence and evolution of state capacity for green industrial policy in the United States, its functions, and current developments.

Read article →

Strategic objectives underpin national climate pledges

2026-08-13 · DOI 10.1038/S41467-026-76467-7

Taryn Fransen; Jonas Meckling

Abstract. Abstract Emissions-reduction pledges under the Paris Agreement are commonly understood as expressions of domestic preferences. Recent scholarship, however, suggests that governments may also use climate pledges strategically to shape the behavior of other political actors. Here, we provide an empirical account of these strategic objectives based on in-depth interviews with 40 policymakers and experts. We identify six key strategic objectives: attracting international climate finance, setting standards for others, enhancing status, preserving credibility, facilitating domestic action, and pre-committing future governments. International actors emerge as the primary target audience. These objectives have distinct implications for ambition. While most encourage stronger pledges, concerns about credibility can constrain ambition. In an exploratory analysis, we also examine how strategic objectives vary across countries. The findings suggest that higher emitters and states with lower levels of democratic institutionalization place greater emphasis on long-term credibility, while countries eligible for climate finance seek to attract it regardless of income or vulnerability. Our findings have implications for ratcheting up international climate commitments.

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The choice-wide behavioral association study: data-driven identification of interpretable behavioral components

2026-08-13 · DOI 10.1038/S41467-026-76681-3

David B. Kastner; Nicole O. Yokota; Christina Y. Lee; Cristofer Holobetz; Ali Chaudhry; Greer Williams; Jane N. Ton; Mihika Mehra; Viktor Kharazia; Dilworth Y. Parkinson; Joseph P. Romano; Peter Dayan

Abstract. Abstract Behavior contains rich structure across many timescales. However, there is a dearth of methods to identify relevant components, especially over the longer periods required for learning and decision-making. Inspired by the goals and techniques of genome-wide association studies, we present a data-driven method-the choice-wide behavioral association study: CBAS-that systematically identifies such behavioral features. CBAS extracts sequences of actions or choices that either significantly differ between groups or correlate with a covariate of interest, using a powerful resampling-based method to correct for multiple comparisons. We illustrate CBAS through application to tasks performed by flies, rats, and humans, showing that it provides unexpected and interpretable information in each case.

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Proceedings of the National Academy of Sciences

The effectiveness of green place-based industrial policy: Evidence from the Inflation Reduction Act

2026-08-14 · DOI 10.1073/PNAS.2600276123

Joep Keuzenkamp; Jacopo Mazza; Bob Rijkers; Katherine Stapleton

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Abstract. Governments across the globe are turning to place-based industrial policies to make the green transition more equitable and politically palatable. The United States' Inflation Reduction Act exemplifies this approach through its "Energy Community" provisions, which offer additional tax incentives for clean energy projects in regions reliant on fossil fuels. We use spatially granular data on renewable energy investment, employment, and voting, and document that these incentives substantially increased local solar investment, but we do not find significant impacts on wind investment, job creation, or political preferences. Estimated abatement costs are well below the social cost of carbon and lower than those of other major U.S. climate policies. The results suggest that place-based green industrial policies can effectively direct clean energy investment toward disadvantaged regions, but their capacity to generate broader local economic or political benefits may be

limited.

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Tariff policy and why economics needs intergroup psychology

2026-08-12 · DOI 10.1073/PNAS.2514670123

Samuel G. Pearson; Matthew J. Hornsey; S. Alexander Haslam

School of Psychology, University of Queensland; University of Queensland Business School

Abstract. Abstract unavailable.

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Mitigating China's food-related water scarcity and spatial inequality via integrated basin-city management

2026-08-11 · DOI 10.1073/PNAS.2610855123

Jinliang Liu; Minghao Zhuang; Changbo Wang; Meili Feng; Donglan Zha; Hao Li; Xiurong Hu; Yanan Ren; Mingyue Pang; Qunwei Wang; Lixiao Zhang

College of Economics and Management & Research Center for Soft Energy Science, Nanjing University of Aeronautics and Astronautics; Humanities and Social Sciences Laboratory of Jiangsu Province for Digital Intelligence Management and Low-carbon Operations for Manufacturing System, Nanjing University of Aeronautics and Astronautics; Key Laboratory of Three Gorges Reservoir Region's Eco-Environment, Ministry of Education, Chongqing University; School of Geographical Sciences, Faculty of Science and Engineering, University of Nottingham Ningbo China; State Key Joint Laboratory of Environmental Simulation and Pollution Control, School of Environment, Beijing Normal University; State Key Laboratory of Regional and Urban Ecology, Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences

Abstract. Food production is constrained by water availability at the basin scale, whereas the governance of food trade primarily operates at the city scale. This spatial mismatch hinders the sustainable development of both food and water systems. To bridge this gap, we develop an integrated framework that combines hydrological (basin) and administrative (city) scales and apply it to China to quantify how virtual water flows embedded in food trade affect inequality in water allocation across and within basins. Our results show that 38% of scarce water is traded across basins, exacerbating interbasin inequality by 6%. In contrast, approximately 30% of scarce water is embodied in intrabasin trade, which alleviates intrabasin inequality. Single-scale interventions, whether through production or consumption adjustments, involve trade-offs among interbasin equality, intrabasin equality, and total water savings. By contrast, an integrated basin-city management strategy at a 50% intervention intensity reduces interbasin inequality by 7% and achieves water savings equivalent to more than four-fifths (80%) of total water use in the Yangtze River basin. It also outperforms existing province-city vertical management models across scales. These findings provide valuable insights for developing global virtual water management strategies that integrate both hydrological and administrative perspectives.

Read article →

The rise of large language models and the direction and impact of US federal research funding

2026-08-11 · DOI 10.1073/PNAS.2601439123

Yifan Qian; Zhe Wen; Alexander C. Furnas; Yue Bai; Erzhuo Shao; Dashun Wang

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Abstract. Federal research funding shapes the direction, diversity, and impact of the US scientific enterprise. Large language models (LLMs) are rapidly diffusing into scientific practice, holding substantial promise while raising widespread concerns.

Despite growing attention to AI use in scientific writing and evaluation, little is known about how the rise of LLMs is reshaping the public funding landscape. Here, we examine LLM involvement at key stages of the federal funding pipeline by combining two complementary data sources: confidential NSF and NIH proposal submissions from two large US R1 universities, including funded, unfunded, and pending proposals, and the full population of publicly released NSF and NIH awards. We find that LLM use rises sharply beginning in 2023 and exhibits a bimodal distribution, indicating a clear split between minimal and substantive use. Across both private submissions and public awards, higher LLM involvement is consistently associated with lower semantic distinctiveness, positioning projects closer to recently funded work within the same agency. The consequences of this shift are agency-dependent. LLM use is positively associated with proposal success and higher early-stage publication output at NIH, whereas no comparable associations are observed at NSF. Notably, the productivity gains at NIH are concentrated in nonhit papers rather than the most highly cited work. Together, these findings provide large-scale evidence that the rise of LLMs is reshaping how scientific ideas are positioned, selected, and translated into publicly funded research, with implications for portfolio governance, research diversity, and the long-run impact of science.

Read article →

A budding science of illegal markets

2026-08-10 · DOI 10.1073/PNAS.2511780123

Jonathan P. Caulkins

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Abstract. Illegal markets create untold societal and environmental harm, but they have attracted surprisingly little scholarly attention. This paper explores illegal markets as a class of objects. It considers similarities and differences across markets, with particular attention to nine: human smuggling, commercial sex, firearms, money laundering, illegal fishing, international wildlife trafficking, counterfeit goods, drugs, and tobacco products. These nine markets display enormous variation, stemming in part from differences in the "physics" of the good or service they provide. Other differences can be seen as adaptations to different policy and enforcement pressures. Some markets have changed in fundamental ways over recent decades, while others remain more stable. The overall rate of true innovation is not necessarily high, but most appear quick to adopt innovations that are created by the larger economy such as cryptocurrencies, secure communications, and dark web distribution. Low rates of investment in research and development, branding, and marketing along with simultaneous high rates of adaptability may be a natural outgrowth of these markets being dominated by relatively small organizations. It is hypothesized that one can group the common illegal markets into those that: 1) Supply goods stolen from a common heritage, either natural (illegal fishing, wildlife trafficking, illegal mining) or cultural (antiquities); 2) Supply goods or services to people committing crimes beyond just consuming the banned products (guns, money laundering); 3) Undermine an authority's rights (smuggling humans across borders, counterfeiting goods, tax-evading cigarettes); and 4) Supply products that harm the user and/or seller (drugs, commercial sex, gambling, raw milk).

Read article →

A descriptive analysis for the markets for counterfeit products

2026-08-10 · DOI 10.1073/PNAS.2522322123

Jay P. Kennedy

Philip Morris International

Abstract. Product counterfeiting is a pervasive crime that involves the unauthorized use of a legitimate trademark, a violation of intellectual property rights. Counterfeiting is also one of many forms of global illicit trade, mirroring legitimate markets in structure and reach. Counterfeit goods span nearly every product category from luxury items and electronics to pharmaceuticals and military components, posing risks that range from consumer deception to threats to public health. The globalization of commerce and trade and the rise of e-commerce and social media have significantly expanded counterfeiters' reach, enabling them to exploit legitimate supply chains and marketplaces and reach consumers around the world. The

production and distribution of counterfeit goods often involve sophisticated networks, including legitimate businesses and transnational criminal organizations, and are sometimes linked to human trafficking and forced labor. Despite regulatory efforts and public-private partnerships aimed at curbing counterfeiting, enforcement remains challenging due to the complexity of global supply chains and the difficulty in distinguishing counterfeit operations from legitimate ones. This article describes what product counterfeiting is and addresses how markets for counterfeit goods represent an illicit parallel market that is in many ways intermingled with aspects of legitimate markets. The role of consumers as drivers of demand for counterfeits is addressed, and motivations for consumers' willful purchase of counterfeit goods are explored. This article also provides exploratory insights about the supply chains for some of these products

Read article →

Americans favor predistributive over redistributive economic policies

2026-08-10 · DOI 10.1073/PNAS.2617191123

David Broska; Jonne Kamphorst; Robb Willer

Center for Socio- Political Data and Center for European Studies and Comparative Politics, Sciences Po; Department of Sociology, Stanford University

Abstract. Redistributive policies designed to reduce economic inequality through taxes and transfers (e.g., safety-net benefits funded by progressive taxation) receive less support in the United States than in many Western democracies. Recent work suggests Americans may be more receptive to "predistributive" policies, which seek to reduce economic inequality before taxes and transfers (e.g., minimum-wage increases). Yet existing evidence leaves open whether this support gap characterizes contemporary Americans overall, and whether it persists when the same policy proposals are described in predistributive vs. redistributive terms, with their core provisions held constant. Across 31 national U.S. survey waves from 2015 to 2024 (total N = 384 , 248), we find greater support for policies classified as predistributive than redistributive. We then fielded a preregistered experiment with a national U.S. sample (N = 1 , 009), randomly varying whether otherwise identical policy proposals were described as increasing earnings for people lower in the income distribution, shifting resources from higher- to lower-income individuals, or presented only with the policy description given in all conditions. Support was higher under predistributive than redistributive descriptions. Comparison with the control condition clarifies this difference: Predistributive descriptions received support comparable to control, whereas redistributive descriptions received lower support. Exploratory analyses suggest perceived respect for hard work and beneficiary deservingness were associated with the support gap. Together, these results suggest Americans prefer policies that reduce inequality by influencing earnings more than through taxes and transfers.

Read article →

Casting light on the workings of illegal fishing markets

2026-08-10 · DOI 10.1073/PNAS.2512081123

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Department of Geography and School of Public Policy and Global Affairs, University of British Columbia; Department of Geography, University of British Columbia; Institute for the Oceans and Fisheries and School of Public Policy and Global Affairs, University of British Columbia

Abstract. About 3.5 million fishing vessels ply the world's oceans, providing livelihoods and food for millions of people. Sound management is key to fisheries' sustainability, but illegal fishing, along with fishing overcapacity generally and environmental change, are putting many fisheries at risk. Illegal fishing is a globally pervasive yet unevenly distributed phenomenon, manifesting in diverse forms across both small-scale and industrial-scale fleets, the latter accounting for about 82 to 93% of the 8.4 to 15.4 million metric tons global annual illicit marine wild fish seafood trade valued at US$6.2 to 12.2 billion. The massive rise in fishing effort since 1950 has not been matched by adequate regulations and enforcement, especially for industrial-scale fishing in the waters of developing countries. About two-thirds of the illicit trade by value originates from fishing grounds in West Africa (27%), East Asia (24%), and Southeast Asia (16%). Remote sensing and AI technologies are

improving detection of illegal fishing, but enforcement remains an issue given its high costs and the relative ease of laundering illegal catch through transshipments at sea, poorly regulated ports, and fraudulent documentation along seafood supply chains. Illegal fishing markets require multilayered policy responses grounded in transparency, international cooperation, and socioeconomic justice. Proposed measures, ranging from mandatory electronic vessel identification and subsidy reform to artisanal fishery protections and stronger port controls, warrant further research into their implementation and effectiveness.

Read article →

Historical slavery predicts contemporary disparities in mortality between Black and White Americans

2026-08-10 · DOI 10.1073/PNAS.2508037123

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Abstract. We tested whether contemporary Black-White mortality disparities in the United States may be rooted in the historical legacy of slavery and perpetuated through structural inequalities. Linking county-level data on the proportion of enslaved people in 1860 with structural inequality indicators from 1990 to 2000 and mortality outcomes from 2010 to 2020, we find that historical slavery is robustly associated with contemporary racial mortality gaps. Our model estimates ~22 additional deaths per 100,000 among Black Americans relative to White Americans for every 10% increase in the 1860 enslaved population. Path analysis models showed that segregation, economic inequality, and racial disparities in poverty and education statistically mediated this effect. We test the plausibility of a causal pathway using an instrumental variable analysis and explore the robustness of our theoretical model in multiple supplemental analyses. Our analyses held even after controlling for a comprehensive suite of historical and contemporary covariates. These findings are consistent with a model wherein slavery's legacy continues to shape present-day Black-White racial health disparities via enduring structural inequities.

Read article →

How illegal wildlife trade adheres to and defies conventional market behavior

2026-08-10 · DOI 10.1073/PNAS.2525056123

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Abstract. Illegal wildlife markets comprise a complex global economy generating criminal profits while harming biodiversity and livelihoods. The negative externalities from biodiversity loss, ecosystem destabilization, and zoonotic disease risk may be orders of magnitude larger than the multibillion dollar annual total market value. Despite decades of international policy attention, fundamental economic characteristics of these markets remain poorly understood. We suggest distinguishing three categories of illegally traded wildlife products based on product durability and intended use: luxury durable goods, nondurable products processed to be food or medicine, and live or perishable exotic pets and plants. Through case studies of elephant ivory, pangolin scales, and succulent plants, we characterize market organization, pricing structures, supply network architecture, and regulatory vulnerabilities across product types. Our analysis suggests illegal wildlife markets exhibit economic characteristics distinguishing them from other illegal markets, including biological production constraints creating absolute supply limits, opportunistic supply networks exploiting existing legal trade infrastructure, and extreme markup structures leaving harvesters with a fraction of final retail value. We conclude by proposing the concept of Accelerating Demand for Novel or eXclusive Assets, or AD NOXA, which describes how scarcity may paradoxically increase rather than decrease the appeal of consuming certain wildlife markets. This paradox creates feedback loops that may drive species toward extinction, despite-or because of-protective regulations and fundamentally limits the effectiveness of supply-side

interventions. Policy responses may be more effective when they integrate supply-side enforcement with demand reduction strategies tailored to the distinct characteristics of illegal wildlife markets.

Read article →

Illegal firearms markets: Transaction patterns, evolving sources, and importance in criminal misuse

2026-08-10 · DOI 10.1073/PNAS.2509754122

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RAND Corporation; Sanford School of Public Policy, Duke University

Abstract. In the United States, the legal status of the 30 to 40 million annual firearms transactions is defined by state and federal regulations. These regulations primarily aim to reduce criminal misuse while preserving access for the large majority of adults who lack a serious criminal record or are otherwise proscribed from possession. Illegal firearms transactions are embedded in the "formal market"-sales, mostly of new guns, by licensed retail dealers required to screen purchasers and maintain records-and the "informal market," where any of the 80 million gun owners may act as a source. A growing fraction of illegal transactions involve privately made firearms, fabricated from commercial kits or additive manufacturing techniques. The firearms used by proscribed people are diverted from the legal supply chain, in some cases by professional intermediaries (thieves, straw purchasers, scofflaw dealers, home fabricators). Subsequent redistribution may involve professional brokers and traffickers. Most such enterprises are small in scale, with organized criminal groups serving primarily as customers rather than distributors. Compared with the drug trade, the illegal gun market is smaller and more exposed to law enforcement. For proscribed buyers, transaction costs are highly differentiated, varying based on the local prevalence of guns, state regulations and enforcement, and individual connections. While preventing dangerous people from obtaining guns is intrinsically difficult in a regime designed for broad accessibility to most adults, regulations can substantially affect illegal commerce, and some regulations reduce criminal use.

Read article →

Illegal markets for cigarettes and e-cigarettes

2026-08-10 · DOI 10.1073/PNAS.2512084122

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Abstract. This paper discusses the long-standing illegal markets for cigarettes and the more recently emerged illegal markets for e-cigarettes. We identify three distinctive market fundamentals that help explain many features of the illegal markets. First, in contrast to some illegal goods, many countries have sizable legal markets for cigarettes and e-cigarettes alongside the illegal markets. In 2024, tobacco products generated almost $1 trillion in revenues; e-cigarettes generated $26 billion dollars. It is estimated that illegal markets account for about 10 percent of global cigarette consumption. Second, the illegal markets have arisen to evade high excise taxes that increase prices and regulations that restrict the availability of desirable features of legal products. The taxes and regulations have been adopted as public health measures with the goal of reducing tobacco consumption; however, these same policies drive the formation of illegal markets. Third, demand-side linkages mean that taxation, regulatory, and enforcement policies targeted at legal and illegal markets for cigarettes will affect legal and illegal markets for e-cigarettes, and vice versa. The connection between the markets means regulatory policy for one product can drive consumers both to the illegal market for the product and to the legal and illegal market for the other product.

Read article →

The illegal market for sex in the United States: Insights from economic research

2026-08-10 · DOI 10.1073/PNAS.2512375122

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Abstract. The illegal market for sex in the United States generates significant economic activity yet remains understudied due to its illegality and clandestine nature. This paper synthesizes existing empirical evidence documenting aspects of market structure, impacts of regulation, and notable changes over time. Criminalization, while intended to suppress the market, often exacerbates harms by displacing activity underground, increasing violence against sex workers, and elevating disease transmission risks. Natural experiments, including decriminalization episodes in Rhode Island and policy shifts in Europe, reveal that decriminalization can reduce sexually transmitted infections and violence against women. Additionally, technological shifts-particularly the rise of online platforms due to the internet-have reshaped market organization. These findings challenge conventional legal and moral frameworks and underscore the need for evidence-based policymaking to guide effective regulation of sex markets.

Read article →

The market for human smuggling: Structure, mechanisms, and paradoxes

2026-08-10 · DOI 10.1073/PNAS.2511019122

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Abstract. Human smuggling is the facilitation of unlawful cross-border movement. It is primarily a commercial activity, offered as a service to willing, paying customers, and a large-scale market catering to millions seeking to circumvent mobility restrictions. Contrary to prevailing media and political narratives that depict smuggling as dominated by powerful and highly sophisticated criminal organizations, this study documents a fragmented, decentralized market composed of independent, localized actors operating through flexible, often temporary networks. These flexible arrangements, coupled with low barriers to entry, confer resilience and adaptability to the market, making enforcement difficult and resource-intensive. Intensified enforcement often produces unintended consequences, including heightened migrant vulnerability and increased reliance on smugglers. Despite regional differences, smuggling operations exhibit striking structural similarities shaped by local knowledge and monitoring costs. Such enduring structural characteristics exist alongside a growing role played by communication technologies, which are reshaping dynamics between service providers and customers. Ultimately, human smuggling remains a market fraught with paradoxes: Migrants face severe risks yet view smugglers as essential facilitators, while states confront the complex task of balancing border enforcement with humanitarian obligations and protections of vulnerable individuals.

Read article →

The policy construction of illegal markets: Exploring variation, success, failure, and unintended consequences

2026-08-10 · DOI 10.1073/PNAS.2515300123

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Abstract. Illegal markets, broadly defined to include markets where only specific transactions are illegal, are the consequence of policy, either a prohibition or highly restrictive regulation. They are generated by a variety of concerns, some moral, some pragmatic. They are often tied to legal markets, complicating the policy process. These markets are ubiquitous, rarely well controlled and little studied as markets. This article examines variations in illegality from the case studies in this Special Feature and synthesizes what can be learned about the success and failure of different approaches to controlling these markets. Control is often weak because powerful legal interests are opposed to tight regulation and, especially in markets with large underlying demand, rarely leads to substantial reductions in the targeted activity, unless draconian methods are used. As demonstrated by drug markets, tough enforcement can have large adverse collateral consequences.

The paper concludes by suggesting that an extension of the harm reduction framework developed specifically for drug policy can be applied usefully to making policy choices in controlling other illegal markets.

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Science Advances

AI agents can coordinate via majority-following beyond human scale

2026-08-14 · DOI 10.1126/SCIADV.AEA6091

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Abstract. Large language models (LLMs) are increasingly deployed in collaborative tasks forming "AI agent societies" where agents interact and influence one another. Whether such groups can spontaneously coordinate without external influence, a hallmark of self-organized regulation in human societies, remains an open question. Here, we use principles from complexity and behavioral science to investigate coordination in AI agent groups through majority-following, a fundamental mechanism for spontaneous consensus formation. Using binary opinion dynamics experiments across multiple LLM architectures and group sizes, we find that agents exhibit majority-following characterized by a universal functional form with a single parameter, the "majority force." This majority force diminishes as group size increases, leading to a critical size beyond which coordination becomes unattainable. The critical group size grows rapidly with model capabilities and, for advanced LLMs, exceeds 1000 agents, larger than typical human informal groups. Our findings have implications for designing collaborative AI systems where coordination could be beneficial or pose safety threats.

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Nature

Undocumented migration does not raise crime rates, huge study finds

2026-08-14 · DOI 10.1038/D41586-026-02542-0

Miryam Naddaf

Abstract. Abstract unavailable.

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Agentic profiles for effective AI governance

2026-08-12 · DOI 10.1038/S41586-026-10805-Z

Atoosa Kasirzadeh; Iason Gabriel

Abstract. Abstract unavailable.

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Towards an equitable future of global photovoltaic waste recycling

2026-08-12 · DOI 10.1038/S41586-026-10905-W

Chen Wang; Jian Zuo; Xinyu Chen; Ruidong Chang; Pengfei Yuan; Kuishuang Feng; Yu Xin; Xi Liu; Peipei Tian; Jing Li; John Laurence Esguerra; Jiashuo Li

Abstract. The world is confronting an escalating crisis of burgeoning photovoltaic (PV) waste1. However, the effectiveness and scalability of prevailing PV waste management approaches remain unclear owing to considerable heterogeneity across regions and over time. Here we develop a comprehensive framework to evaluate the economic and climate benefits of local versus outsourced recycling, covering mainstream technologies. Considering supply-side material constraints, global PV waste will reach 297-402 million tonnes by 2060, with middle-income regions such as China becoming major contributors after 2040. Notwithstanding anticipated technological advancements, the break-even point for global PV waste recycling remains more than a decade away. Combining region-specific recycling technologies with outsourced recycling strategies yields the maximal global net benefits, reducing greenhouse gas emissions by up to 3.32 billion tonnes of CO2-equivalent and generating cumulative net benefits of US$529.1-935.5 billion by 2060. However, outsourced recycling raises inequality concerns for low-income regions. Our results show that a well-designed declining-subsidy scheme effectively mitigates these inequalities, particularly in the early stages. We suggest that regionally adapted recycling strategies and international cooperation, with a focus on technology transfer and funding for recycling capacity in low-income regions, provide effective ways to achieve equitable and scalable PV waste circularity.

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NIH limits funding for research on the health effects of public policy

2026-08-11 · DOI 10.1038/D41586-026-02489-2

Max Kozlov

Abstract. Abstract unavailable.

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Science

A democracy tax in support of science

2026-08-13 · DOI 10.1126/SCIENCE.AEL2482

John Palfrey

John Palfrey is president of the John D. and Catherine T. MacArthur Foundation, Chicago, IL, USA.

Abstract. Over the past 19 months, the devastating effects of federal cuts to science funding in the United States have become plain: Young researchers are leaving scientific fields. Labs are struggling to fund promising projects, limiting hiring, training, and sometimes shuttering altogether. Data collected over decades will instead collect dust, ending future scientific discovery and innovation. But perhaps the most serious consequence has been overlooked-the importance of free scientific inquiry to democracy.

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Previous Week Catch-Up

Political Communication

The Knowledge Gap in a High-Choice Media Environment: Experimental Evidence from Online Search

2026-08-06 · DOI 10.1080/10584609.2026.2708131

Roberto Ulloa; Leonard Tiedemann; Peter Selb; Celina Kacperski

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Abstract. Persistent inequalities in political knowledge are a central concern in political communication. We organize the mechanisms underlying the knowledge-gap literature by distinguishing between individual preconditions, structural features of the information environment, and topic characteristics. Within this framework, we note that self-directed information seeking, a prototypical form of intentional exposure, has received little attention despite its importance in navigating today's complex information environment. We conducted a field experiment in Germany combining randomized encouragements and passive browser tracking to examine how individuals with varying education levels acquire policy-specific knowledge through online search. Participants were randomly assigned to one of three conditions (verbal encouragement, financial encouragement, or control) to seek information on three salient policy topics differing in divisiveness and complexity (child support, energy transition, and cannabis legalization). We estimate both intention-to-treat (ITT) and local average treatment effects (LATE) of information seeking on post-search knowledge outcomes, with a focus on education and civic knowledge as moderators. While the interventions equalized information-seeking behavior, the results provide some support for the knowledge gap hypothesis: knowledge gains were concentrated among participants with higher education or baseline civic knowledge, who, according to our post-hoc exploratory analyses, appeared more effective at navigating search results. These findings indicate that a narrowing of knowledge inequalities goes beyond motivation: it calls for both individual-level interventions to strengthen citizens'skills and structural-level adaptations to foster more equitable learning environments.

Read article →

Journal of Communication

"Communicating in the Air Force": the National Society for the Study of Communication, the Journal of Communication , and U.S. military education, 1950-1962

2026-08-05 · DOI 10.1093/JOC/JQAG012

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Abstract. Abstract This article explores the intertwined history of the National Society for the Study of Communication (NSSC), its Journal of Communication, and U.S. Air Force educational programs. Founded in 1950 (and renamed the International Communication Association in 1969), NSSC developed a close partnership with the Air Force's Air University in its early days. The growth of the academic society and the expansion of communication courses in military education programs were then mutually reinforcing: NSSC leaders-including several Air Force officers and civilian consultants-helped define and refine military communication curricula. Air University resources supported NSSC projects, most notably the Journal of Communication. Drawing on extensive archival research and a close reading of the Journal's content during its first decade, this article argues that these collaborations not only nurtured NSSC's institutional development, but also shaped a specific conception of communication emphasizing vocational training, efficiency, and the transmission of command.

Read article →

Through their eyes: neural and behavioral evidence for enhanced immersion via point-of-view editing

2026-08-04 · DOI 10.1093/JOC/JQAG011

Moran Ovadia; Shiri Davidovitch; Romi Sion; Naama Shoham; Nimrod Keren; Sergei Freedman; Cynthia Cabanas; Dov Rubinstein; Mark A Thornton; Jonathan Frome; Liad Mudrik; Tim J Smith; Gal Raz

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Tel Aviv University , Tel-Aviv,

Abstract. Abstract Point-of-view (POV) editing-the juxtaposition of a character's gaze with the object of that gaze-is presumed to function as a unique empathic device that dissolves viewer-character boundaries; however, direct empirical evidence remains scarce. This study provides the first systematic functional magnetic resonance imaging (fMRI) and behavioral investigation into whether POV editing facilitates shared attention and intentionality. 39 adults viewed 8 original narrative movies containing POV sequences or closely matched control conditions. POV sequences increased fixation durations on characters' faces and enhanced subsequent recognition of gazed-at content. Neurally, POV editing elicited stronger responses in regions implicated in mentalizing and increased shared, item-specific activation patterns across viewers within areas associated with perspective-taking and goal-directed control. Multivariate decoding further indicates that a character's gaze induced more robust neural signatures of character identity on subsequent target elements. Together, these findings support a functional role for POV editing in promoting joint attention and shared intentionality in audiovisual messages.

Read article →

Communication Research

Twitter Conversations: Gendered Issue Publics Among Women

2026-08-03 · DOI 10.1177/00936502261470086

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Abstract. We investigate how gender shapes women's political discussion on Twitter using an issue publics framework. We test whether the conditions for gendered issue publics to form among women around feminine issues exist using data on political discussion via social media in the 2020 and 2022 elections. Our data includes political discussion among political candidates, elites, and non-elites, or regular people. Our results reveal several key findings. First, we find that men elites were more likely to discuss feminine issues, such as health care, compared to women elites, but we found no gender differences between regular Twitter users in our sample. Democratic women elites are more likely to discuss equity-based feminine issues, such as equal pay, than Republican women elites, but we find no partisan differences among non-elite women. We also find no gender or partisan differences at the elite and non-elite levels in the discussion of trait-based feminine issues, such as education. Finally, we show that in 2022, the Dobbs decision that rescinded the constitutional right to an abortion in the U.S. increased the salience of this issue among both women and men. Overall, our research suggests that the conditions for issue publics among women may not strongly exist as we find that non-elite men talk about feminine issues too.

Read article →

Political Analysis

Let Them Eat Pie: Addressing Sample Selection in Multiparty Elections

2026-08-05 · DOI 10.1017/PAN.2026.10049

Ali Kagalwala; Thiago M. Q. Moreira; Guy D. Whitten; Yongzhi Xu

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Abstract. Abstract Elections are central to the study of politics. When studying parties' vote shares across districts, scholars are encouraged to use compositional-outcome models in order to test their theories about what factors shape the dynamics of electoral support. Existing compositional modeling approaches incorrectly deal with scenarios in which not all parties compete in every electoral district. Because unobserved factors that affect a party's decisions to contest a district are likely correlated with its performance in districts where the party fielded candidates, failing to account for partial contestation is likely to result

in sample selection bias. Addressing sample selection in a compositional setting is challenging because the outcomes are in log-ratio form, and thus the errors often deviate from normality. To deal with these issues, we introduce a novel maximum likelihood approach which accounts for this type of sample selection and demonstrate through simulations that our method outperforms commonly used solutions, including the conventional Heckman correction. We illustrate the utility of our approach by analyzing the 2017 and 2019 U.K. parliamentary elections in English constituencies.

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Nature Human Behaviour

Individual and differential harm in redistricting

2026-08-06 · DOI 10.1038/S41562-026-02511-7

Cory McCartan; Christopher T. Kenny

Abstract. Social scientists have developed dozens of measures for assessing partisan bias in redistricting. However, these measures are not easily adapted to other groups, including groups defined by race, class or geography, nor are they applicable to single- or no-party contexts, such as local redistricting. Here we propose a unified framework of harm for evaluating the impacts of a districting plan on individual voters and the groups to which they belong, to overcome these limitations. We consider a voter harmed if their chosen candidate is not elected under the current plan but would be under a different plan. Harm improves on existing measures by both focusing on the choices of individual voters and directly incorporating counterfactual plans. We discuss strategies for estimating harm using redistricting simulations, and demonstrate the utility of our framework through three applications to US redistricting. Overall, harm provides a flexible way to precisely quantify redistricting's individual impacts. McCartan and Kenny develop a framework for quantifying the individual-level impacts of redistricting in the USA, showing its applicability for identifying partisan gerrymandering and differential effects across populations.

Read article →

Longitudinal effect of early social media use on standardized learning outcomes during school career

2026-08-03 · DOI 10.1038/S41562-026-02522-4

Marco Gui; Chiara Respi; Giovanni Abbiati; Cristiana Paladini; Sofia Ercolanoni; Francesca Milzani; Tiziana Pirola; Giovanni Vezzoli

Abstract. Abstract This study uses difference-in-differences models to evaluate the impact of early social media use on academic performance among children aged 7-16 years. A dataset comprising 5,227 students from northern Italy was constructed by merging data from a questionnaire on lifetime digital media habits with longitudinal standardized test scores in Italian, English and mathematics. Participants who created a personal social media account in 6th, 7th or 8th grade were separately matched with late adopters who initiated their first social media account in the 9th grade or later. The findings provide compelling evidence that early engagement with social media adversely affects learning outcomes. Additional findings on the effects of early social media use on other academic outcomes for which we do not have longitudinal data are in line with our main findings. We also find supporting evidence that a substantial part of this negative effect is mediated by students' smartphone pervasiveness in key moments of the day.

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Nature Communications

Human learning of probability distributions is biased toward moderate structural complexity

2026-08-08 · DOI 10.1038/S41467-026-76247-3

Tianyuan Teng; Li Kevin Wenliang; Hang Zhang

Abstract. Abstract Inferring hidden environmental structures, which commonly involves learning arbitrary probability distributions from limited samples, is essential to optimal and adaptive behaviors across various cognitive domains. However, it remains largely unknown how the internal representations constructed by humans may deviate from actual probabilistic structures, and what computational processes, operated under inherent cognitive limitations, give rise to these representations. We first develop a structured distribution report task to reveal human participants' internal representations, with findings verified in a further distribution recognition task. Across eight behavioral experiments (including one pre-registered study) in two modalities, participants estimate the overall probability density reasonably well, but exhibit a systematic bias toward moderate structural complexity, reporting too many clusters when the true distribution is a single Gaussian and too few when it contains many clusters. We then build a series of learning models in the framework of approximate Bayesian inference fit to our behavioral data. Through model comparisons, we reconstruct the prior beliefs guiding the evolution of participants' internal representations. The best-fitting model for human reports reduces structure growth rate as complexity increases, effectively constraining the complexity of internal representations within memory limitations.

Read article →

Strategic timing for the decarbonization contributions by China's photovoltaic manufacturers in global energy transition

2026-08-06 · DOI 10.1038/S41467-026-76424-4

Qingyang Lin; Chenglin Li; Kai Wang; Kai Fang; Meixuan Xin; Siqi Wang; Anqi Xu; Zhenci Xu; Yue Lu; Xiang Gao

Abstract. Abstract Photovoltaic (PV) technology is central to clean electricity and the Sustainable Development Goals. However, China's PV manufacturing industry, which accounts for nearly two-thirds of global capacity, is often criticized for its high carbon emissions. Existing studies typically assign emission responsibility to manufacturers while attributing decarbonization benefits to power generation, overlooking upstream producers' contributions. Here we propose a Producer Decarbonization Contribution framework to quantify these long-overlooked contributions at national and global scales. We show that China's PV manufacturers contribute 31.4%-73.8% of total PV decarbonization in China and 27.9%-45.4% globally. We further identify a strategic timing, defined as the point at which the PV's decarbonization contributions reach their maximum, occurring around 2035 in China under the Announced Pledges Scenario and 2030 globally under the Net Zero Emissions Scenario proposed by the International Energy Agency. These findings suggest maintaining supportive policies for China before 2035, while adopting more stringent measures globally after 2030 to avoid overproduction. The proposed framework that enables adaptive industrial policy design can be extended to other renewable energy technologies.

Read article →

Collective problem decomposition improves the wisdom of deliberative crowds

2026-08-05 · DOI 10.1038/S41467-026-76365-Y

Federico Barrera-Lemarchand; Victoria Lescano-Charreau; Julieta Ruiz; Nuria Caceres; Facundo Carrillo; Mariano Sigman; Joaquin Navajas

Abstract. Abstract Understanding when and why social interaction improves human judgment is a central question in the behavioural sciences. We examine whether collective accuracy improves when groups break complex estimation problems into simpler components and generate approximate intermediate estimates, a reasoning strategy we call Collective Fermi Estimation. Across three studies analysing 1140 online group deliberations in text chatrooms we study its causal effects and linguistic signatures. In Study 1 (N = 500), spontaneous use of problem decomposition predicts lower collective estimation error. Study 2 (N = 240) provides causal evidence: groups instructed to apply problem decomposition outperform groups instructed to combine their initial guesses. Study 3 (N = 160) shows that the benefits are larger when the strategy is applied collectively rather than individually. Altogether, we show that problem decomposition constitutes a key mechanism behind the wisdom of deliberative crowds and we provide tools to detect and promote it.

Read article →

Global trade-offs between consumption, carbon prices and equity

2026-08-05 · DOI 10.1038/S41467-026-75815-X

Mengyu Li; Jarmo S. Kikstra; Lorenz Keyer; Jason Hickel; Keywan Riahi; Manfred Lenzen

Abstract. As the remaining carbon budget continues to dwindle, economic growth strains efforts to limit global warming to 1.5 C. Reducing unnecessary production and consumption in high-income regions and among the wealthy, and organising production more around human needs and human well-being, may alleviate pressures on technology-driven decarbonisation strategies. We employ a modified integrated assessment model to explore the interaction between converging per-capita consumption, climate policy stringency and equity. Our pathways demonstrate that lower-consumption futures are associated with substantially lower warming, 0.3-1.1 C across different carbon price trajectories, relative to high-consumption futures. Comparatively, global convergence of incomes between regions has a small influence on global temperature outcomes (<0.1 C). These findings indicate that reducing high per-capita consumption in high-income regions is a primary lever for achieving futures with equitable and decent living standards while meeting global climate targets. While such transformations may face political challenges, our scenarios suggest that unconstrained growth futures make it considerably harder to remain within safe climate limits. The authors use a modified integrated assessment model to show that lower-consumption and more equitable futures could substantially reduce warming, highlighting the importance of reducing high consumption in affluent regions to meet climate targets.

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Proceedings of the National Academy of Sciences

Implicit biases change more slowly in larger cities

2026-08-06 · DOI 10.1073/PNAS.2514189123

Andrew J. Stier

The Santa Fe Institute

Abstract. Implicit biases remain a contributor to discrimination despite widespread endorsement of equality. Although there has been a large-scale reduction in sexuality, race, and skin tone biases in the past two decades, there is a current shortage of quantitative theories that describe how and why biases change. Here, I present a theory of implicit bias change, which focuses on changes in exposure due to cultural information and how the speed of these changes varies geographically. This model makes counterintuitive predictions when compared with previous findings, including 1) that biases change more slowly in larger cities and 2) that one-off interventions are less effective in larger cities. I confirm these predictions using 14 million tests of implicit biases from U.S. residents over 14 y and experimental data from 1,150 individuals. Documenting the typical timescales and patterns that govern implicit bias change is an important step toward designing interventions that can facilitate further reductions in biases and prevent biases from increasing.

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Oxytocin increases trust in men low in dispositional trust

2026-08-06 · DOI 10.1073/PNAS.2602655123

Bodo Vogt; Paul Bengart; Carolyn Declerck; Ernst Fehr

Chair in Empirical Economics and Health Economics, Faculty of Economics and Management, Otto von Guericke University Magdeburg; Faculty of Business and Economics, University of Antwerp; Institute of Social Medicine and Health Systems Research, Otto von Guericke University Magdeburg; Zurich Center for Neuroeconomics, Department of Economics, University of Zurich

Abstract. In recent years, increasing skepticism regarding oxytocin's (OT) influence on social behavior arose. Low power, HARKing (hypothesizing after the results are known), and replication failures have clouded the field. Here, we directly address these concerns with a high-powered, preregistered study that offers robust evidence for a causal effect of OT on trust among males with a low disposition to trust. We recruited 359 low-trusting men who participated in a trust game with complete anonymity. Results show that OT administration significantly increased trusting behavior by roughly 15%, with consistent effects across regression models with and without controls for personality traits. A pooled data analysis incorporating a previous sample (n = 219) of low-trusting individuals further strengthens this conclusion, yielding a statistically significant 16.9% increase in trust. No interaction effect was found between OT and the degree of dispositional trust, suggesting OT's effect may be uniform across the low-trusting spectrum. These findings present a strong case for OT's selective trust-enhancing role. By isolating OT's impact within a well-defined subpopulation and experimental context, this study resolves a key question in the debate over neurobiological determinants of trust.

Read article →

Claims of sex-dependent effects in the behavioral and brain sciences are infrequently supported by valid statistical comparisons

2026-08-04 · DOI 10.1073/PNAS.2608703123

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Abstract. Interest in precision medicine has accelerated efforts to identify sex-dependent responses to treatments and interventions. Those efforts have, both historically and today, been concentrated in fields related to the brain and behavior. Here, we show that even though articles claiming sex-dependent effects are far more prevalent in the neurosciences than in other fields, those claims are infrequently supported by appropriate statistical evidence. Of 200 recent articles in the behavioral and brain sciences with a claim of a sex- or gender-dependent effect in the title, only 24% supported the claim with appropriate evidence-that is, they compared the effect statistically across sex and reported results consistent with the claim. In 9%, an appropriate test was described but the results were incomplete or missing; another 9% contained results that directly contradicted the claim. In most articles (58%), the effect was not compared statistically across sex. Across six brain-related research areas (behavioral sciences, clinical neurology, neurosciences, psychiatry, psychology, and substance abuse), neurosciences had the lowest rate of appropriate evidence (18%) and psychology the highest (39%). Studies on nonhuman animals supported the claim significantly less frequently (15%) than studies on human participants (34%; P = 0.002). The use of appropriate analytical approaches was unrelated to journal rank or the citation impact of the article. We conclude that claims of sex- or gender-dependent effects in the behavioral and brain sciences are often not supported by appropriate evidence, revealing a widespread inferential gap that may undermine translation of sex-dependent findings into evidence-based treatments.

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Optimizing infectious disease mitigation under dynamic conditions

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Abstract. Mitigation measures are essential for controlling the spread of infectious diseases during pandemics and epidemics, but they impose considerable societal, individual, and economic costs. We developed a general framework that combines simulation of disease dynamics with optimal control to determine mitigation strategies that balance infection and

mitigation costs. Optimizing this trade-off, we identified three surprising effects: first, assuming a constant reproduction number R 0 , the optimal response is typically "all-or-nothing": depending on disease severity, either strict mitigation or none at all is optimal, with intermediate levels emerging only in restricted regimes that we characterize analytically. Second, under seasonal variations, optimal mitigation is stricter during winter. Interestingly, a single wave of infections still arises in spring, replacing the autumn/winter waves known for classical influenza. Third, during steady vaccination campaigns, even optimal mitigation can result in transient infection waves. Finally, we quantify the cost of delayed mitigation onset and show that even short delays can substantially increase total costs-if the disease is severe. Overall, our framework is easily applicable to general and complex settings and thereby presents a versatile tool to explore optimal mitigation strategies for endemic and pandemic infectious disease.

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Science Advances

Assessing pedestrian responses to autonomous and personal mobility robots in crowded public spaces

2026-08-07 · DOI 10.1126/SCIADV.AEF2576

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Abstract. Robots increasingly share spaces with people, supporting delivery services and mobility for the aging populations, yet their ability to share space comfortably lacks understanding and benchmarks for designers and policy-makers. We compared human-robot (HRI) and human-human (HHI) interactions across four real-world crowd datasets spanning Europe, North America, and Asia, using a unified pipeline to detect interactions, stratify by crowd density, and model pedestrian behavior. Local motion patterns (speed, acceleration, and jerk) remained closely matched between HRI and HHI across all densities. In contrast, proxemics diverged, with effects that grew approximately linearly with robot speed and weakened under higher crowding: In the dataset with the faster navigating robot, pedestrians maintained about 0.23 meters more clearance around the robot than around other pedestrians under less crowded conditions and about 0.05 meters more under more crowded conditions, while in the dataset with the predominantly stationary robot, the corresponding differences were small and inconsistent across crowding levels. The main conclusions were robust to parameter variations and remained stable across a broad range of motion-processing and interaction-labeling settings. Our findings provide density- and speed-aware benchmarks for proxemics in social robot navigation and empirically grounded targets for design, evaluation, and modeling across robotics and urban mobility.

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Escalation dynamics and the severity of wars

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Abstract. Although very large wars remain an enduring threat in global politics, we lack a clear understanding of how some wars become large and costly, while most do not. There are three possibilities: Large conflicts begin with intense fighting, accumulate severity over a long duration, or escalate in intensity over time. Using detailed within-conflict data on civil and interstate wars (1946-2008), we show that escalation dynamics-variations in fighting intensity within an armed conflict-are the primary mechanism for producing large conflicts. Civil wars, however, tend to deescalate once they become very large,

limiting their overall severity, while interstate wars exhibit a persistent risk of continual escalation. Simple models of within-conflict severity dynamics show that this distinction compactly explains the historical size distributions of civil and interstate wars, and unconditioned escalation is a plausible mechanism for Richardson's law-the power law pattern in the frequency and severity of interstate conflicts. The dynamics of escalation within conflicts have broad implications for conflict theory and risk assessment.

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Higher-order trade-offs in hypergraph community detection

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Abstract. Extending community detection from pairwise networks to hypergraphs introduces fundamental theoretical challenges. Hypergraphs exhibit structural heterogeneity with no direct graph analog: Hyperedges of varying orders can connect nodes across communities in diverse configurations. Crucially, this means that any hypergraph community detection algorithm must make a choice regarding which order or balance of hyperedges it prefers to maintain or split during partitioning. We address these challenges by developing a unified framework for community detection in nonuniform hypergraphs under the hypergraph stochastic block model. We introduce a general signal-to-noise ratio that enables a quantitative analysis of trade-offs unique to higher-order networks, such as which hypergedges we choose to split across communities and how we choose to split them. Building on this framework, we derive a Bethe Hessian operator for nonuniform hypergraphs that provides efficient spectral clustering with principled model selection. We characterize the resulting spectral detectability threshold and compare it to belief propagation limits, showing the methods coincide for uniform hypergraphs but diverge in nonuniform settings. Synthetic experiments confirm our analytical predictions and reveal systematic biases toward preserving higher-order and balanced shape hyperedges. Application to empirical data demonstrates the practical relevance of these higher-order detectability trade-offs in real-world systems.

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Steppe productivity decline contributed to smallpox outbreak and the collapse of the last nomadic empire in the 1750s

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Abstract. The collapse of the Dzungar Khanate in the 1750s marked the end of the world's last nomadic empire and reshaped Eurasian geopolitics. However, whether environmental and societal factors interplayed and contributed to its collapse remains unclear. Here, we reconstructed steppe productivity variability across the Dzungar heartland since 1625 using 441 tree-ring series based on an extreme growth ratio. Our record shows that the Khanate's collapse coincided with the lowest steppe productivity in the past four centuries during 1751-1761. The weakening societal resilience approaching a tipping point intensified the effects of the rapid steppe productivity decline, which cascaded into famine, conflict, and trade

breakdown. Smallpox then reemerged and escalated into a catastrophic epidemic, transforming the environmental crisis into the collapse of the Dzungar Khanate. Our findings underscore the urgent need to strengthen pastoral resilience and epidemic surveillance in steppe regions as climate stress intensifies and cross-border epidemic threats escalate.

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Nature

How scientists can tackle the paradox of political polarization

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Peter Calow

Abstract. Abstract unavailable.

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Science

How should India approach AI?

2026-08-06 · DOI 10.1126/SCIENCE.AEL0463

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Abstract. In an unstable geopolitical environment, countries are looking to secure access to artificial intelligence (AI) capabilities, which are seen as essential to economic growth, scientific research, and national security. Few have placed sovereignty as squarely at the center of their AI strategy as India-and few are as well positioned to achieve it. But to do so, it needs to look beyond infrastructure-chips, data centers, and large language models-and pay more attention to the application layer above them: AI agents, the software and tools that accomplish real-world work. If the effort succeeds, India may pioneer a template for genuine AI sovereignty that other nations can follow.

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The bridge between science and policy: How IPBES redefined global conservation policy and its path to societal impact

2026-08-06 · DOI 10.1126/SCIENCE.AEK0207

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Abstract. Abstract unavailable.

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The climate benefits of retiring a fully operational internal combustion engine vehicle

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Abstract. Internal combustion engine (ICE) vehicles represent the single largest source of carbon dioxide emissions for most US households. Although battery electric vehicles (BEVs) considerably reduce use-phase emissions, the transition poses a fundamental life cycle optimization challenge: Is there a net climate benefit to retiring a functional ICE vehicle before its end of life? Retiring a functioning ICE vehicle incurs a fundamental trade-off between BEV manufacturing emissions and decreased vehicle operation emissions. In this study, we evaluated this trade-off across a comprehensive range of retirement scenarios and found that scrappage-and-replacement yields consistent emissions reductions across most contexts, including the retirement of new ICE vehicles. Although scrapping a functional asset remains economically prohibitive under current market conditions, these results demonstrate a major mitigation potential for policy interventions, such as enhanced scrappage subsidies, to address the emissions of an increasingly durable ICE fleet.

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